Chapter 1 Determinants

The determinant of a matrix is a scalar value computed from its elements.

Definition: the determinant of an n by n matrix A is defined by the Leibniz formula.

det A = sum over permutations sign(p) times product of a i p(i)

Motivation: determinants tell us when a system of linear equations has a unique solution.

1.1 Geometric Meaning

The absolute value of the determinant equals the volume of the parallelepiped.

Theorem: a square matrix is invertible if and only if its determinant is non zero.

Proof: if the determinant is non zero the reduced row echelon form is the identity.

Figure 1-1 shows the area interpretation for two by two matrices.

Chapter 2 Matrices

A matrix is a rectangular array of numbers arranged in rows and columns.

Definition: the identity matrix has ones on the diagonal and zeros elsewhere.

Theorem: the product of two invertible matrices is invertible.